Examining the Separability of Co-Occurring ERP Components in a Single Condition Using Time-Frequency Analysis

In ERP research, it is relatively common for a single anomaly to cause distinct effects in two different components; e.g., plausible but unpredictable nouns can elicit both an N400 effect and a frontal positivity effect (DeLong et al., 2014; Kuperberg et al., 2020). When two effects co-occur in the same condition, it is often assumed that both effects are present in every trial in the anomalous condition; however, because ERP data are averaged across many trials and participants, this assumption is not entirely justified. It could be that some trials elicit only the first component strongly and other trials elicit only the second component strongly, and that averaging them together gives the illusion that the components co-occur more weakly. There is some evidence of this at the person level, with participants in one study showing different effects for the same anomaly in a single time window (Caffarra et al., 2019; Tanner & Van Hell, 2014). If two components co-occur in individual trials, it would imply that *both* cognitive processes associated with these components are being taxed more heavily by the anomaly; contrastingly, if the components do not co-occur on a trial-by-trial basis, this might imply a taxing of one process *or* the other, depending on how a comprehender interprets a given trial's anomaly.

To test the possibility that co-occurring effects may be separable, time-frequency information—typically considered "noise" by ERP analyses, disappearing in the process of averaging—can be used to determine which effects are present in a given trial. Hagoort et al. (2004) demonstrated that N400 effects to world-knowledge and semantic violations that had near-identical peaks and scalp distributions were very different when examined via Time-Frequency Analysis (TFA); world-knowledge violations showed moderate activity in both the gamma and theta frequency bands during the N400 time window, while semantic violations showed only strong activity in the theta frequency band. This difference implies either different neural generators or different activity in the same generator, which further implies different cognitive processes being taxed by the different anomaly types. Clearly, TFA offers insight when examining the differences between conditions. However, time-frequency data is available at the trial-level as well; if comparing *between* conditions is possible, it is plausible that sorting trials using time-frequency data *within* a single condition would allow researchers to determine whether "co-occurring" effects are truly co-occurring.

The present study examined plausible but unpredictable critical words in English (e.g., "Bruno is [watching/editing] a movie in his garage. He realizes he forgot the **popcorn**…") and demonstrated both an N400 effect ($p < .001$) in clusters Cz, C4, Pz, and P4 (Fig. 1) and a frontal positivity effect ($p = .003$) in clusters F3, Fz, C3, and Cz. By one recent account (Hansen et al., 2025), the co-occurrence of the N400 and frontal positivity could indicate that there was an interacting top-down/bottom-up conflict between the incorrect predicted meaning and the incoming orthographic information; however, this interpretation requires that the effects are truly co-occurring. To test whether this is the case, time-frequency data can be used to sort whether each trial exhibits just a frontal positivity effect, just an N400 effect, both effects, or neither.

The analysis is currently being conducted; the relative distribution of trials and the resulting mixed effects model will reveal whether the effects are mostly co-occurring or mostly separate. If the effects are mostly co-occurring, this would provide support for the prior explanation by Hansen et al. (2025). Alternatively, if the effects are mostly separate, this would indicate that they may not be the result of said conflict, but represent participants reacting to similar anomalies using different cognitive processes. Regardless of which result is found, examining trial-level time-frequency data could be a highly informative tool for EEG and MEG researchers who find co-occurring effects, and could help shed light on existing conflicts around the interpretation of late ERP components in language comprehension.

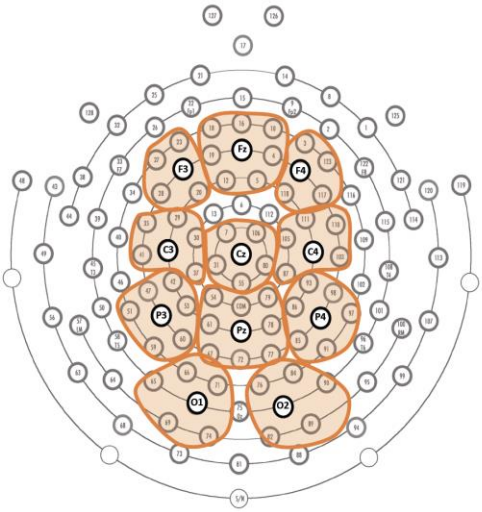

Figure 1. Map of cluster distributions for the experiment, with nose oriented up. As is somewhat typical, the N400 effect was observed in clusters Cz, C4, Pz, and P4 (right-posterior) and the frontal positivity effect was observed in clusters F3, Fz, C3, and Cz (left-anterior).

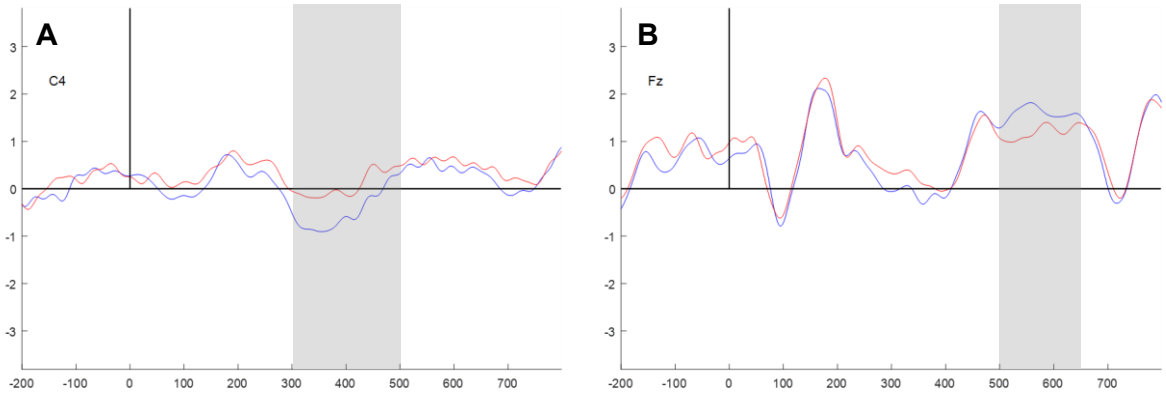

Figure 2. Representative clusters for the N400 effect (A) and P600 effect (B). The *baseline* condition is in red and the *incongruent* condition is in blue. The appropriate time windows have been highlighted in gray.

References

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